

The Microstructure of Wealth Transfer in Prediction Markets

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Abstract

We analyze 72.1 million transactions executed on Kalshi, a CFTC-regulated prediction market, from June 2021 through November 2025. Aggregate prices are well calibrated, but calibration conceals a persistent internal transfer. Because YES and NO positions at the same cost basis risk identical capital for identical payoffs, neither risk preferences nor misperception of the probability level can price them differently; in the data, YES contracts underperform NO contracts of equivalent cost basis by up to 64 percentage points, concentrated at longshot prices. Demand responds to the affirmative framing of the event itself; we term the premium this flow pays the *optimism tax*. Decomposing returns by order-book role identifies who collects it: liquidity takers earn mean excess returns of -1.12% per trade while makers earn $+1.12\%$, and maker returns are statistically and economically indistinguishable across position directions, indicating accommodation of biased flow rather than superior forecasting. An exact decomposition of taker losses attributes roughly three-quarters of the transfer to execution within price-direction cells and the balance to the accommodation of directionally tilted flow; selection across price levels, the longshot bias proper, contributes essentially nothing. The transfer ranges from 0.17 percentage points in financial contracts to more than 7 in media and geopolitical categories, and it emerged only after post-election volume growth attracted professional liquidity providers: the maker-taker gap reversed by 5.3 percentage points. The forecasting accuracy of prediction markets derives not from the rationality of individual participants but from a microstructure in which liquidity provision monetizes behaviorally biased order flow.

Keywords: prediction markets, market microstructure, optimism tax, longshot bias

JEL Classification: D91, G13, G14

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1 Introduction

Slot machines on the Las Vegas Strip return approximately 93 cents per dollar wagered. On Kalshi, a prediction market regulated by the Commodity Futures Trading Commission (CFTC), traders have committed substantial capital to longshot contracts with realized returns as low as 43 cents per dollar. Prediction markets are zero-sum, so these losses accrue to other participants; this paper asks to whom, and by what mechanism.

Binary event contracts make the question tractable in a way that equity markets do not. A contract pays \$1 if a specified event occurs and expires worthless otherwise, so price maps directly to implied probability, calibration can be measured against resolved outcomes, and every dollar lost by one side of a trade is collected by the other. The setting therefore permits an unusually direct test of informational efficiency (Fama, 1970): if some population of traders systematically overpays, the data reveal both the overpayment and the population that collects it.

Using 72.1 million trades comprising \$18.26 billion in notional volume from Kalshi's launch in June 2021 through November 2025, we find that aggregate prices are well calibrated: contracts priced at p cents win p percent of the time, to within one percentage point on average and 2.4% at worst across the 99 price levels. Calibration at the market level, however, conceals two sharp divisions beneath it. The first is by contract direction: YES positions priced at 1 cent return -41% on average, while NO positions at the same price return $+23\%$, a divergence of 64 percentage points between two positions with identical prices and identical payoff structures. The second is by order-book role: liquidity takers, who cross the spread to execute against resting orders, earn -1.12% per trade, while liquidity makers, who post those orders, earn $+1.12\%$.

The direction asymmetry does more than document a mispricing; it discriminates among explanations of the favorite-longshot bias, the century-old tendency of bettors to overpay for low-probability outcomes (Griffith, 1949; Thaler and Ziemba, 1988). Whether the bias reflects risk-loving preferences or systematic misperception of small probabilities remains contested (Snowberg and Wolfers, 2010). Binary contracts permit a test that wagering markets do not: a YES position and a NO position at the same cost basis risk identical capital for identical payoffs at identical implied probabilities. Preferences over payoffs therefore cannot price them differently, and misperceptions that depend only on the probability level

apply to both sides alike. The 64-percentage-point gap between them isolates a third channel: demand responds to the affirmative framing of the event itself. Takers disproportionately buy YES at longshot prices, precisely where such contracts are most severely overpriced, and incur losses larger than the longshot bias alone implies. We term the premium this flow pays the *optimism tax*.

Decomposing returns by order-book role identifies who collects the tax. Takers earn negative excess returns at 80 of 99 price levels, and makers earn positive excess returns at the same 80. Maker returns are statistically and economically indistinguishable across the directions of the positions they hold, which rules out superior forecasting as the source of their profits: makers do not pick winners, they profit by standing on the passive side of a flow that is predictably biased. An exact decomposition of taker losses makes the accounting precise: selection across price levels, the longshot bias proper, contributes essentially nothing; the directional tilt of taker flow priced against the YES/NO wedge contributes roughly one quarter; and execution within price-direction cells contributes three-quarters. The transfer also varies in ways that implicate participant behavior rather than market design. It ranges from 0.17 percentage points in financial contracts, a category operating near full efficiency, to more than 7 percentage points in media and geopolitical categories, a forty-fold spread that plausible differences in spreads alone cannot generate.

The transfer is historically contingent, and its emergence identifies a necessary condition. From launch through 2023 the sign was reversed: takers earned +2.0% per trade against amateur liquidity suppliers, even though the longshot bias was present throughout. The maker-taker gap swung by 5.3 percentage points around the 2024 election, when a surge in volume made systematic liquidity provision economically worthwhile and drew professional makers to the platform. Behavioral bias alone does not produce a wealth transfer; it is a latent resource that requires sophisticated counterparties, at sufficient scale, to harvest.

This paper makes three contributions. First, we identify the favorite-longshot bias on Kalshi as a framing phenomenon: the payoff-identity test rejects explanations based on preferences over payoffs, and the concentration of the asymmetry in affirmative contracts locates the distortion in how an event is described rather than in the odds it carries. Second, we show that the bias finances a persistent wealth transfer from takers to makers, decompose the transfer exactly into price-selection, direction-selection, and execution channels, and characterize the conditions under which it operates: the transfer requires professional liquid-

ity provision, scales with the behavioral composition of each category’s participants, and is absent where questions demand probabilistic reasoning. Third, we quantify calibration and the longshot bias across the full price spectrum of a regulated prediction market at census scale, providing a benchmark for a rapidly growing venue class now entering the mainstream of U.S. derivatives regulation.

A fast-forming literature examines who profits in prediction markets. [Bartlett and O’Hara \(2026\)](#) adapt equity-market adverse-selection measures to Kalshi and show that optimistic YES demand cross-subsidizes the adverse selection that makers bear; our payoff-identity test complements their equilibrium analysis by isolating framing as the source of that optimism. [Bürge, Deng and Whelan \(2025\)](#) model Kalshi’s maker–taker economics with demand that is symmetric in YES and NO; the asymmetry we document is a moment such models cannot match. [Deleep, Lee, Bai, Suresh and Dhawan \(2026\)](#) report overpayment for affirmative outcomes in mention markets; we show the pattern is market-wide, quantify it across all 99 price levels, and tie it to the order-book role of the traders who pay it. On Polymarket, [Reichenbach and Walther \(2025\)](#) document YES-side overtrading as a flow phenomenon, [Akey, Grégoire, Harvie and Martineau \(2026\)](#) show that profits concentrate among limit-order makers, and [Della Vedova \(2026\)](#) decompose wallet-level profits into forecasting and execution components, finding that execution rather than forecasting separates winners from losers. Our results give the venue-level counterpart of that conclusion on a regulated exchange: the profitable side of the market is defined by how it trades, not by what it knows. The retail losses we document extend a lineage running from [Barber and Odean \(2000\)](#) in equities and [Moskowitz \(2021\)](#) in sports betting to the newest retail venue class.

The paper proceeds as follows. Section 2 describes prediction markets and Kalshi. Section 3 describes the data and measurement. Section 4 establishes calibration and the longshot bias. Section 5 presents the YES/NO asymmetry and the framing test. Section 6 traces the resulting transfer across order-book roles, categories, and time. Section 7 interprets the mechanism and states the limits of the evidence. Section 8 concludes.

2 Prediction Markets and Kalshi

Prediction markets are exchanges in which participants trade binary contracts referencing real-world events ([Wolfers and Zitzewitz, 2004](#)). A contract settles at \$1 if the referenced event occurs and at \$0 otherwise, and trades beforehand at prices between 1 and 99 cents.

Because every open contract pairs a YES position against a NO position, the market is strictly zero-sum gross of fees: in contrast to equity markets, where aggregate returns reflect underlying cash flows, prediction markets generate no external return, and gains and losses net to zero across participants.

Kalshi launched in June 2021 as the first prediction market to operate as a CFTC-designated contract market. Early listings concentrated on macroeconomic data releases and weather outcomes, and activity remained modest for the platform’s first three years: quarterly volume first exceeded \$50 million in mid-2023 (Figure 1). Growth accelerated sharply in late 2024. In October of that year, the U.S. Court of Appeals for the D.C. Circuit affirmed Kalshi’s right to list contracts on the control of Congress ([U.S. Court of Appeals for the D.C. Circuit, 2024](#)), and the ensuing presidential election drove quarterly volume from \$74 million in 2024 Q3 to \$1.7 billion in 2024 Q4. The introduction of sports contracts in early 2025 sustained the expansion: quarterly volume reached \$4.5 billion in 2025 Q3 and \$8.8 billion in the partial final quarter of the sample.¹

Volume is heavily concentrated by topic (Figure 2). Sports accounts for 73% of notional volume over the sample period, followed by politics at 14% and cryptocurrency at 5%; no other category, including the financial contracts that later serve as an efficiency benchmark, exceeds 5%. This composition matters for the aggregate results that follow, because the magnitude of the maker–taker transfer varies substantially across categories (Section 6).

3 Data and Methodology

The dataset comprises 7.68 million markets and 72.1 million trades collected from Kalshi’s public REST API, spanning the platform’s first trade on June 30, 2021 through November 25, 2025. Each trade records an execution price between 1 and 99 cents, the number of contracts exchanged, a timestamp, and the side (YES or NO) of the liquidity taker. Each market records its resolution outcome and a category classification. The dataset and processing code are publicly available.²

Every executed trade pairs a liquidity *taker*, whose marketable order crossed the spread, against a *maker*, whose resting limit order it consumed, and the two receive opposite positions: if the taker bought YES at 10 cents, the maker holds NO at a cost of 90 cents.

¹ Data collection concluded at 17:00 ET on November 25, 2025; 2025 Q4 figures are therefore incomplete.

² <https://github.com/jon-becker/prediction-market-analysis>

Because the feed identifies the taker side of every trade, both positions created by a trade can be assigned a role, and aggregate performance decomposes exactly by role.

We evaluate positions in terms of capital risked. For a trade executed at price p_i cents, the YES position has cost basis $c_i = p_i$ and the NO position has cost basis $c_i = 100 - p_i$. A position’s cost basis is both its capital at risk per contract and, divided by 100, its implied probability of paying out; references to price hereafter denote cost basis, so that YES and NO positions are directly comparable at equal price.

Let $o_i \in \{0, 1\}$ indicate whether position i pays out at resolution. We measure performance with two quantities. The *mispricing* of a set S of positions is the difference between its realized win rate and its mean implied probability,

$$\delta_S = \frac{1}{|S|} \sum_{i \in S} o_i - \frac{1}{|S|} \sum_{i \in S} \frac{c_i}{100}, \quad (1)$$

which should be indistinguishable from zero, for any subset S selected on information available at execution, if prices are efficient. The *gross excess return* of position i is its return on capital risked,

$$r_i = \frac{100 \cdot o_i - c_i}{c_i}, \quad (2)$$

gross of platform fees.³

All calculations use resolved markets only: markets that were voided or delisted, or that remained open at the end of the sample, are excluded, as are markets with less than \$100 in notional volume. Coverage is dense across the price spectrum (Figure 3); the sparsest price decile, 81–90 cents, contains 5.8 million trades.

Trades are not independent observations. Every position in a market shares that market’s single resolution outcome, and markets in the same event family resolve jointly, frequently as mutually exclusive alternatives. The strike prices of one numeric question and the candidates of one race are examples of such families. All standard errors and t -statistics in this paper therefore cluster, either at the market level (554,223 clusters) or at the coarser event-family level (193,341 clusters). The two levels bracket the dependence structure from opposite sides: market-level clustering treats markets of the same event as independent and so ignores the

³ Kalshi assesses trading fees on marketable orders at execution, while resting orders trade without fee on most markets. Accounting for fees would therefore lower taker returns and leave maker returns largely unchanged, widening every gap reported below; gross returns are the conservative choice.

negative correlation between mutually exclusive resolutions, which inflates standard errors, while event-family clustering absorbs that dependence. We report t -statistics at both levels for each headline estimate and treat the market level as the conservative benchmark.

4 Calibration and the Longshot Bias

First documented in parimutuel horse-race betting (Griffith, 1949; Thaler and Ziemba, 1988), the favorite–longshot bias is the tendency of bettors to overpay for low-probability outcomes; related mispricings persist in modern sports betting markets (Moskowitz, 2021). Whether the bias reflects risk-loving preferences or systematic misperception of small probabilities remains contested (Snowberg and Wolfers, 2010). In a well-calibrated market, positions priced at c cents pay out approximately c percent of the time; under the bias, low-priced positions win less often than their price implies while high-priced positions win more often.

Kalshi exhibits this pattern (Figure 4). Positions priced at 5 cents pay out 4.18% of the time ($n = 1,407,530$), a mispricing of $\delta_S = -0.82$ percentage points in the sense of Equation (1) that corresponds to a mean gross excess return of -16.4% . Positions priced at 95 cents pay out 95.83% of the time. Deviations elsewhere are individually small, but they are uniformly signed at the tails: every price below 20 cents underperforms its implied probability, and every price above 80 cents outperforms.

Viewed as a forecasting instrument, the market performs well: realized frequencies track implied probabilities to within one percentage point on average, within two at 91 of the 99 price levels, and within 2.4 everywhere. But the market is zero-sum, so even small systematic deviations are not forecast errors that wash out; they are transfers. If longshot positions are persistently overpriced, whoever holds them persistently pays whoever supplies them. The remainder of the paper identifies what kind of error produces the overpricing (Section 5) and the two sides of the transfer it finances (Section 6).

5 The YES/NO Asymmetry

What kind of error produces the longshot bias? Explanations divide into accounts based on risk-loving preferences and accounts based on probability misperception (Snowberg and Wolfers, 2010). Wagering markets struggle to separate the two because every longshot bet couples a small probability to a large payoff; any test of the probability channel is simulta-

neously a test of the payoff channel. Binary event contracts break this coupling. A YES position and a NO position at the same cost basis risk identical capital for identical payoffs at identical implied probabilities; the only difference between them is which description of the world the holder is betting on. Any account in which demand depends only on the payoff distribution, whether through preferences over payoffs or through misperception of the probability level, requires the two positions to earn the same return. This section shows that they do not.

5.1 The Asymmetry at Equivalent Prices

Nothing in the structure of a binary contract distinguishes a YES position from a NO position: both risk c cents to win $100 - c$, and under efficiency both should earn expected returns near zero at every price. The data contradict this symmetry (Figure 5). YES positions priced at 1 cent return -41% on average, while NO positions at the same price return $+23\%$: a divergence of 64 percentage points between two positions with identical prices and identical payoff structures.

The NO advantage is pervasive. NO outperforms YES at 69 of 99 price levels (Cohen’s $d = 0.27$), with the advantage concentrated at the extremes: NO earns higher returns at every price from 1 to 10 cents ($d = 1.09$) and at every price from 91 to 99 cents ($d = 1.11$). On a dollar-weighted basis, YES positions return -1.02% and NO positions $+0.83\%$, a gap of 1.85 percentage points driven by the systematic overpricing of affirmative contracts. Under clustered inference (Section 3), the per-position gap of 1.30 percentage points carries $t = 3.2$ with event-family clusters and $t = 1.2$ with market clusters, and holding price fixed the gap is 1.16 percentage points ($t = 2.8$ and 1.1 , respectively); the dollar-weighted gap is estimated far less precisely ($t = 2.1$ and 0.5), because dollar weights concentrate its variance in a handful of very large markets.

The asymmetry bears directly on the source of the longshot bias. Because YES and NO positions at the same price share an identical payoff structure, preferences over payoffs cannot generate the divergence documented here, and misperceptions that depend only on the probability level apply to both sides alike; only an account in which the affirmative framing of an event distorts demand is consistent with the data.

5.2 Takers Prefer Affirmative Bets

The overpricing of affirmative longshots is sustained by the composition of order flow. Figure 6 decomposes the capital at risk at each price into four components by role and direction. At longshot prices, the composition is dominated by taker demand for the affirmative side: taker-held YES accounts for 41–47% of capital priced at 1–10 cents, while maker-held YES accounts for only 20–24%. The high-priced end of the spectrum shows the mirror image: 43% of capital priced at 99 cents is maker-held NO, against 23% taker-held NO. The two tails are mechanically linked, since every taker purchase of YES at 1 cent creates a maker NO position at 99 cents, so the figure describes a single phenomenon viewed from both sides: takers demand affirmative longshots, and makers absorb the negative favorites that result. We term the premium this flow pays the *optimism tax*. Whether the makers who absorb the flow profit through directional judgment or through accommodation is a question about the other side of the market, which Section 6 takes up.

5.3 Demand Framing or Question Framing?

One alternative deserves statement. YES-at-5-cents and NO-at-5-cents positions live in different markets, and the exchange chooses which side of each question is the affirmative one. If question writers systematically phrase salient, narrative-rich outcomes as YES, then affirmative longshots would attract biased demand partly because of how the exchange selects and words its questions, not only because of how traders respond to a given wording. The distinction does not rescue efficiency, and it remains a framing account: either way, returns depend on the description of the event rather than on its probability or payoff, which no rational benchmark permits. But it locates the frame partly on the supply side, and it qualifies the interpretation of the asymmetry as a pure trader-psychology parameter. Separating demand framing from question framing requires variation in wording that holds the underlying event fixed, for example the same event listed under different phrasings across venues; we leave this test to future work.

6 The Maker–Taker Wealth Transfer

The asymmetry of Section 5 identifies a systematically mispriced flow. This section identifies who monetizes it, shows that the losses concentrate in a single population defined

by order-book role rather than by any information the winners hold, and decomposes the transfer exactly into its constituent channels.

6.1 Decomposing Returns by Role

Decomposing aggregate returns by order-book role reveals that the two sides of the market fare very differently (Table 1). Across the full sample, takers earn a mean gross excess return of -1.12% per trade while makers earn $+1.12\%$; the maker–taker gap of 2.25 percentage points carries $t = 8.5$ with standard errors clustered by event family and $t = 3.5$ under the more conservative market-level clustering.

The divergence is most pronounced at the tails (Figure 7). At a price of 1 cent, taker positions pay out 0.43% of the time ($n = 1,343,181$) against an implied probability of 1%, a mean excess return of -57% ; maker positions at the same price pay out 1.57% of the time ($n = 996,039$), a mean excess return of $+57\%$. Toward the center of the price range the divergence compresses but does not close: at 50 cents, takers earn -2.65% and makers $+2.66\%$. In all, takers earn negative excess returns at 80 of 99 price levels, and makers earn positive excess returns at the same 80. The aggregate miscalibration of Section 4 is thus not diffuse; it is concentrated in a single population, with takers bearing the losses and makers capturing the gains.

6.2 Spread Compensation or Forecasting?

Two candidate explanations would strip the transfer of its behavioral content. Makers might simply earn the bid–ask spread as compensation for providing liquidity, or they might possess superior directional judgment, knowing when to take the negative side of an overpriced question. The direction decomposition of maker returns weighs against both.

Maker returns depend only modestly on the direction of the position taken (Figure 8). Makers who bought NO outperform makers who bought YES at 59 of 99 price levels; on a volume-weighted basis, makers earn $+0.77\%$ on YES positions and $+1.25\%$ on NO positions, a gap of 0.47 percentage points that is statistically indistinguishable from zero under clustered inference ($t = 1.4$ with event-family clusters, $t = 0.5$ with market clusters). This weighs decisively against superior forecasting: the direction gap is an order of magnitude smaller than the role gap, its sign is not stable across price ranges, with NO outperforming in the 1–10 and 91–99 cent ranges but YES outperforming in the 76–90 cent range, and effect

sizes are negligible throughout (Cohen’s $d \approx 0.02\text{--}0.03$). Makers profit not by predicting which side will win but through a mechanism that operates symmetrically across directions: accommodating a taker flow that is systematically willing to overpay for affirmation.

A direction gap this small rules out forecasting, but it cannot by itself say how the transfer is collected. That requires an exact accounting of where taker losses arise, to which we now turn.

6.3 Decomposing the Transfer into Channels

The claim that makers profit through structure can be made exact. For each price c and direction $d \in \{Y, N\}$, collect every position with cost basis c and direction d , maker- and taker-held alike, into a cell, and let $q_{c,d}$ denote the cell’s realized win rate and $\delta_{c,d} = q_{c,d} - c/100$ its mispricing in the sense of Equation (1). Write $\bar{\delta}_c = \frac{1}{2}(\delta_{c,Y} + \delta_{c,N})$ for the direction-neutral mispricing at price c and $\Delta_c = \delta_{c,Y} - \delta_{c,N}$ for the YES/NO wedge of Section 5. A role $R \in \{T, M\}$ is described by the share $w_{c,d}^R$ of its positions in each cell, from which follow its price distribution $w_c^R = w_{c,Y}^R + w_{c,N}^R$, its directional tilt $s_c^R = w_{c,Y}^R / w_c^R$, and its own within-cell win rates $q_{c,d}^R$. The role’s mean per-trade mispricing then decomposes identically as

$$\delta_R = \underbrace{\sum_c w_c^R \bar{\delta}_c}_{\text{price selection}} + \underbrace{\sum_c w_c^R (s_c^R - \frac{1}{2}) \Delta_c}_{\text{direction selection}} + \underbrace{\sum_{c,d} w_{c,d}^R (q_{c,d}^R - q_{c,d})}_{\text{execution}}. \quad (3)$$

The first term is what the role would earn if it kept its price distribution but chose directions by coin flip and won at pooled rates: it isolates selection into over- or underpriced price levels, the longshot bias proper. The second term prices the role’s directional tilt, interacting its deviation from direction-neutrality at each price with the market-level wedge; it is the optimism tax in the narrow sense, and it compares only positions with identical cost bases. The third term is a within-cell residual: at a fixed price and direction, whether the role’s positions pay out more or less often than the pooled population in the same cell. Effective-spread payments are confined to this term, because a resting order fills on the favorable side of fair value and a marketable order on the unfavorable side even at the same executed price; but the term also absorbs within-cell market selection, so it bounds spread capture from above rather than measuring it. The identity parallels the wallet-level split of per-trade profit into directional and execution components in [Della Vedova \(2026\)](#), but differs

in construction: the counterfactual is the pooled win rate of each price–direction cell rather than a fair-price benchmark estimated from preceding trades, so the decomposition is exact rather than benchmark-dependent, and it separates a third channel, direction selection, that a two-way split cannot see.

Table 2 reports the three channels for taker flow; maker components are equal and opposite. Three features of the full-sample column stand out. First, price selection is negligible: takers lose 0.14 points to the underperformance of their longshot positions (1–20 cents), but their positions at mirror-image favorite prices (81–99 cents) collect an almost exactly offsetting 0.13 points from the same calibration curve, netting -0.02 . The longshot bias, measured per trade in probability units, is not where the money moves. Second, direction selection costs takers 0.24 points. This is the optimism tax operating through prices: a flow tilted toward YES meeting a market in which YES is persistently the overpriced side. Notably, most of it accrues in the middle of the price range, where the wedge is large in probability units and taker flow is most strongly tilted: positions priced 21–80 cents contribute 0.21 of the 0.24. The tail asymmetry of Figure 5, spectacular in return-on-capital units, is a small share of aggregate probability mass. Third, the execution term dominates, at -0.86 points or roughly three-quarters of the total: holding both price and direction fixed, taker positions pay out 0.86 points less often than the pooled population in the same cells. Maker profit is therefore earned almost entirely without directional opinion: approximately one-quarter from accommodating a tilted flow, and three-quarters from standing on the favorable side of executions within cells. This is precisely the structure, rather than the information, that the direction evidence of Section 6.2 implies. The subperiod columns of the table, which split the sample at the start of 2024 Q4, are taken up in Section 7.2.

6.4 Variation Across Categories

If the transfer reflects the exploitation of uninformed demand, categories that attract less sophisticated participants should exhibit larger gaps. Table 3 reports mean excess returns by role for the eight largest categories, ordered by the size of the maker–taker gap.

The variation is large. Finance exhibits a gap of 0.17 percentage points; with takers losing 0.08% per trade, the category operates close to full efficiency. At the other extreme, Media and World Events exhibit gaps exceeding 7 percentage points. Sports, the largest category by volume, shows a moderate gap of 2.23 percentage points, but applied to \$6.1 billion

in taker volume even this moderate rate generates the bulk of the transferred wealth. The ordering is consistent with participant selection: categories whose questions demand probabilistic reasoning attract calibrated traders, while categories that invite casual or emotional participation do not. We return to this interpretation in Section 7.

6.5 Evolution over Time

The transfer is not a fixed feature of the market’s design. In Kalshi’s early years its sign was reversed: from launch through 2023, taker returns averaged +2.0% per trade and maker returns -2.0% (Figure 9). Taker returns first turned negative on a sustained basis in 2024 Q2, and the gap widened sharply around the 2024 election, as sample volume grew from \$37 million in 2024 Q3 to \$819 million in 2024 Q4.⁴ Averaged across quarters, the maker–taker gap swung from -2.9 percentage points before the election to $+2.5$ percentage points after it, a reversal of 5.3 percentage points.

A change in the composition of taker flow cannot account for the reversal. If the transfer had emerged because new participants arrived with stronger longshot preferences, the distribution of taker volume should have shifted toward low-priced contracts. It did not (Figure 10). The share of taker volume in longshot positions (1–20 cents) was essentially flat, 4.8% before the election versus 4.6% after. The distribution instead shifted toward the middle of the price range: the 91–99 cent share fell from 44% of taker volume in 2021 Q3 to 18% in 2025 Q4, while mid-range prices grew correspondingly. Taker behavior, if anything, became less extreme even as taker losses increased.

The natural interpretation is that the wealth transfer requires not only biased taker flow but sophisticated counterparties positioned to monetize it. In the low-volume early period, resting orders were supplied largely by unsophisticated participants who lost to comparatively informed takers. The post-election volume surge made systematic liquidity provision economically worthwhile, and the professional makers it attracted extract value from taker flow across all price levels.

⁴ Volume figures in this subsection refer to the resolved-market sample of Section 3 and are therefore smaller than the exchange totals of Figure 1.

7 Discussion

The results describe a market whose aggregate accuracy coexists with a persistent internal transfer. This section interprets the mechanism behind that transfer, the conditions under which it operates, and the limits of the evidence.

7.1 The Mechanism of Extraction

A central question in any zero-sum market is whether profitable participants win through superior information or superior structure. The evidence here supports structure. If makers won by forecasting, their returns should depend on which side they held; instead, the direction gap of Section 6.2 is an order of magnitude smaller than the role gap and statistically indistinguishable from zero, and the decomposition of Section 6.3 attributes three-quarters of maker profit to within-cell execution and the balance to accommodating a directionally tilted flow, with the longshot bias proper contributing nothing. Makers do not pick winners. They profit by standing on the passive side of a flow that is predictably biased.

Figure 11 illustrates the direction channel at its most visible with a stylized example. A taker, drawn by narrative appeal and the prospect of a large payoff, crosses the spread to buy an affirmative longshot at 5 cents when the event’s probability is 1%. The maker accommodates the order not because it holds a contrary view of the event but because the flow itself is mispriced: the trade transfers 4 cents per contract in expectation regardless of which outcome is realized. Aggregated over the roughly half of longshot volume that taker YES demand represents (Section 5.2), this per-contract edge becomes the optimism tax documented above. Similar accommodation dynamics have been reported on Polymarket (Reichenbach and Walther, 2025; Akey et al., 2026) and Betfair (Whelan, 2025), suggesting that liquidity provision, rather than forecasting skill, is the dominant source of profit across prediction market venues.

7.2 The Professionalization of Liquidity

The temporal evidence of Section 6.5 shows that behavioral bias alone does not produce a wealth transfer. The subperiod columns of Table 2 locate the reversal within the decomposition: of the 2.10-point swing in mean taker mispricing between the two periods, 1.83 points come from the execution channel, 0.27 from direction selection, and less than 0.01

from price selection.⁵ The behavior of the execution term rules out reading it as a structural spread cost. A fixed toll for immediacy cannot be paid *to* the takers who cross the spread, yet the term stood at +0.87 points across the thirteen quarters through 2024 Q3: resting orders sat, on average, on the wrong side of fair value, and consuming them was profitable. From 2024 Q4 the term is -0.97 . What the execution term measures is not the width of the spread but the calibration of whoever supplies the quotes: amateurs before the surge, professionals after it.

The direction channel tells a complementary story. The YES/NO wedge is effectively a constant of the market: weighted by taker flow, it measures -1.14 points before 2024 Q4 and -1.16 after. What changed is the flow that meets it. Takers were direction-neutral in aggregate before the surge, placing 48.6% of trades on the YES side, and strongly tilted after it, at 70.4%. This refines the compositional evidence of Section 6.5: the post-election flow is no more longshot-seeking than before, but it is substantially more *affirmative*-seeking, and that tilt, priced against an unchanged wedge, contributes the 0.27-point balance of the reversal.

What converted the bias into a reliable source of extraction was depth. Systematic liquidity provision carries fixed costs, and at a few million dollars of quarterly volume the biased flow was too small to be worth harvesting. The post-ruling surge changed that calculus, drawing in professional makers whose entry simultaneously compressed taker returns at every price level and captured the premium that amateur makers had previously ceded. The bias, in other words, was a latent resource; sufficient volume made its extraction economic.

7.3 Category Variation and Participant Selection

The forty-fold variation in the maker–taker gap across categories (Table 3) is difficult to reconcile with any explanation rooted in market design, since all categories share the same order book, tick size, and fee schedule. It is readily explained by who chooses to trade.

- **Finance (0.17 pp).** Questions such as “Will the S&P 500 close above 6,000?” demand probabilistic reasoning and attract participants who already think in expected values. Casual bettors have no edge, recognize as much, and self-select out. The category demonstrates that a prediction market can operate near full efficiency.

⁵ Subperiod means are trade-weighted and therefore differ from the quarter-averaged figures of Section 6.5, which weight quarters equally.

- **Politics (1.02 pp).** Political questions carry emotional stakes, but the population that trades them follows polling and has calibrated beliefs across repeated election cycles. Engagement is partisan; reasoning remains broadly probabilistic.
- **Sports (2.23 pp).** The modal category by a wide margin. Sports bettors exhibit well-documented distortions, including team loyalty, recency effects, and attachment to narrative, and a fan buying a championship longshot is purchasing hope rather than expected value. The gap is moderate, but applied to the category’s dominant volume share it accounts for most of the aggregate transfer.
- **Crypto (2.69 pp).** Price-threshold questions invite directional narratives rather than probability estimates, and the participant pool overlaps heavily with retail momentum traders.
- **Entertainment, Media, and World Events (4.79–7.32 pp).** The largest gaps share a common feature: a minimal barrier to perceived expertise. Following celebrity news or reading headlines produces familiarity, and familiarity is easily mistaken for calibration.

The pattern reduces to two variables: the technical barrier to informed participation, and the degree to which a question invites emotional rather than probabilistic reasoning. Where the barrier is high and the framing clinical, the market approaches efficiency; where the barrier is low and the framing narrative, the optimism tax reaches its maximum.

7.4 Limitations

Four limitations qualify these results. First, the data contain no unique trader identifiers, so order-book role serves as a proxy for sophistication. The proxy is standard in the microstructure literature, but it misclassifies sophisticated traders who cross the spread to act on time-sensitive information. Second, historical quotes are unavailable, so the bid–ask spread at execution cannot be observed and spread capture cannot be strictly decoupled from within-cell selection; the execution term of Equation (3) bounds spread capture from above rather than measuring it. Third, the results describe a single U.S.-regulated venue. Offshore platforms with different fee structures, leverage, and participant pools may exhibit different dynamics, and the cross-venue consistency noted above rests on studies with differing

methodologies. Fourth, positions in the same market share a single resolution outcome and markets in the same event family resolve jointly, so trades are far from independent observations; inference therefore clusters at the market or event-family level throughout (Section 3). The maker–taker gap survives this correction comfortably, with t -statistics above 3.4 at either clustering level, and its sign holds across categories, quarters, and price levels. The YES/NO asymmetry is measured less precisely: dollar weighting concentrates roughly half of the clustered variance in the ten largest markets, and a gap that is significant under event-family clustering is only marginal under market-level clustering, so its magnitude carries more uncertainty than its pervasiveness across price levels suggests.

8 Conclusion

The promise of prediction markets is that trading aggregates dispersed information into accurate probabilities. Our analysis of 72.1 million trades on Kalshi shows that the resulting prices are indeed well calibrated, but that the calibration is not produced by uniformly rational participants. Because YES and NO positions at the same cost basis are identical gambles, the 64-percentage-point return gap between them cannot reflect preferences over payoffs or misperception of the odds; it reflects a costly attraction to affirmative framing. The market divides into a taker population that systematically overpays for low-probability affirmative outcomes and a maker population that captures the resulting premium through passive liquidity provision. The transfer between them is governed by composition and maturity rather than by market design: where questions are dry and quantitative, the market operates near efficiency; where they invite tribalism and hope, it functions as a mechanism for transferring wealth from the optimistic to the calculated.

These findings do not diminish prediction markets as forecasting instruments. Prices remain accurate precisely because biased flow is met by capital with an incentive to correct it; the optimism tax is what that correction costs, and who pays it. An accurate probability at the market level and a losing trade at the individual level are, in this setting, the same transaction viewed from different sides.

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Table 1: Mean Gross Excess Return by Order-Book Role

This table reports the mean gross excess return per trade, defined in Equation (2), separately for liquidity takers and liquidity makers across the full resolved-market sample of 72.1 million trades, together with the maker–taker gap. Standard errors, in percentage points, cluster by event family (193,341 clusters) and, more conservatively, by market (554,223 clusters); Section 3 describes the clustering scheme.

Role	Mean excess return	Event-family clusters		Market clusters	
		SE	t	SE	t
Taker	−1.12%	0.13	−8.51	0.32	−3.45
Maker	+1.12%	0.13	+8.53	0.32	+3.46
Maker − Taker	+2.25 pp	0.26	+8.52	0.65	+3.46

Table 2: Decomposition of Mean Taker Mispricing into Extraction Channels

This table reports the decomposition of mean taker per-trade mispricing δ_T into the three channels of Equation (3), in percentage points. Maker components mirror taker components with opposite sign to within rounding. The two subperiod columns split the sample at the start of 2024 Q4; components may not sum to totals due to rounding. Subperiod means are trade-weighted.

Component	Full sample	Through 2024 Q3	From 2024 Q4
Price selection	−0.02	−0.03	−0.02
Direction selection	−0.24	+0.01	−0.26
Execution	−0.86	+0.87	−0.97
Total (δ_T)	−1.12	+0.84	−1.25

Table 3: Mean Gross Excess Return by Order-Book Role and Market Category

This table reports mean gross excess returns by order-book role for the eight largest market categories, ordered by the size of the maker-taker gap. Returns are defined in Equation (2) and computed on the resolved-market sample.

Category	Taker (%)	Maker (%)	Gap (pp)	Trades (M)
Finance	−0.08	+0.08	0.17	4.4
Politics	−0.51	+0.51	1.02	4.9
Sports	−1.11	+1.12	2.23	43.6
Weather	−1.29	+1.29	2.57	4.4
Crypto	−1.34	+1.34	2.69	6.7
Entertainment	−2.40	+2.40	4.79	1.5
Media	−3.64	+3.64	7.28	0.6
World Events	−3.66	+3.66	7.32	0.2

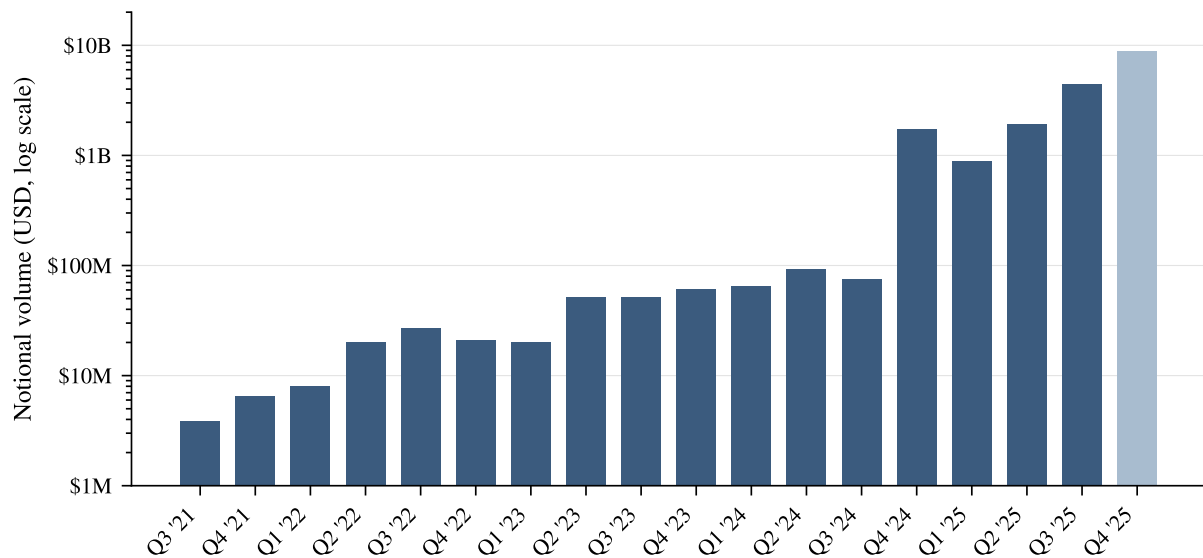


Figure 1: Quarterly Notional Trading Volume on Kalshi

Quarterly notional trading volume on Kalshi (log scale). The final bar (2025 Q4, light shading) reflects a partial quarter; data collection concluded on November 25, 2025.

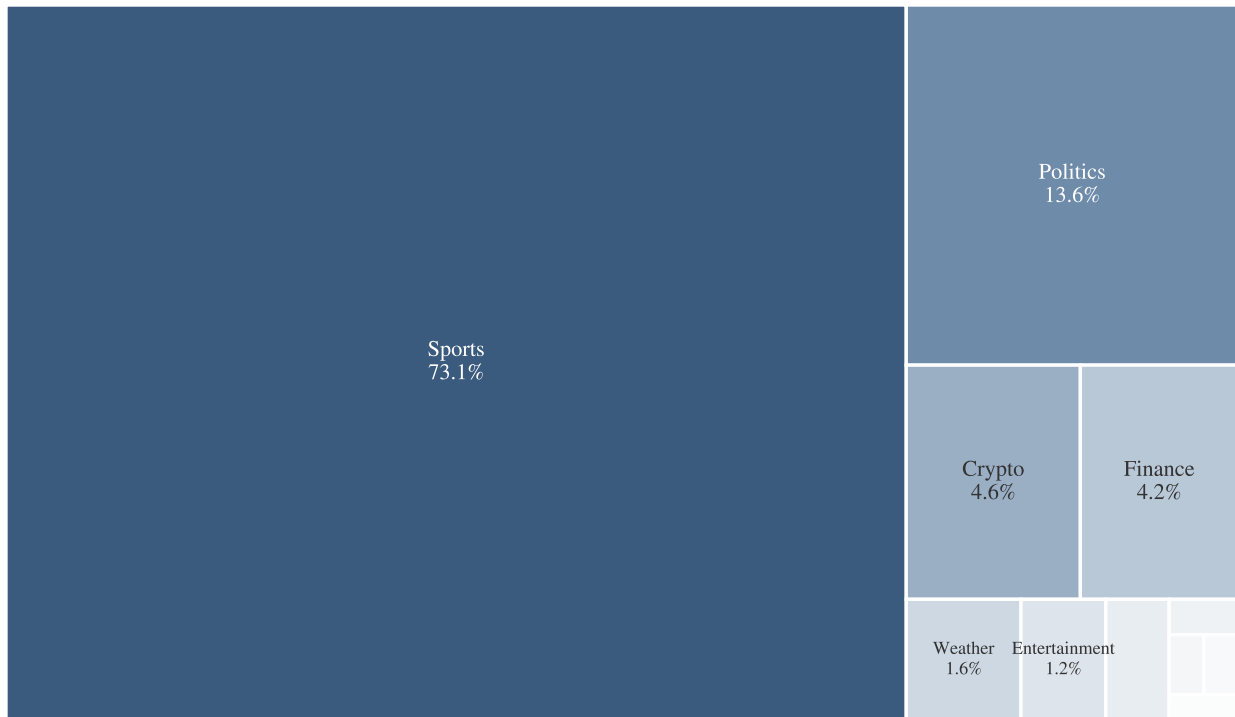


Figure 2: Composition of Notional Volume by Market Category

Cell areas are proportional to notional volume over the sample period. Categories contributing less than 1% of volume (Other, Media, World Events, Esports, and Science/Tech) are unlabeled.

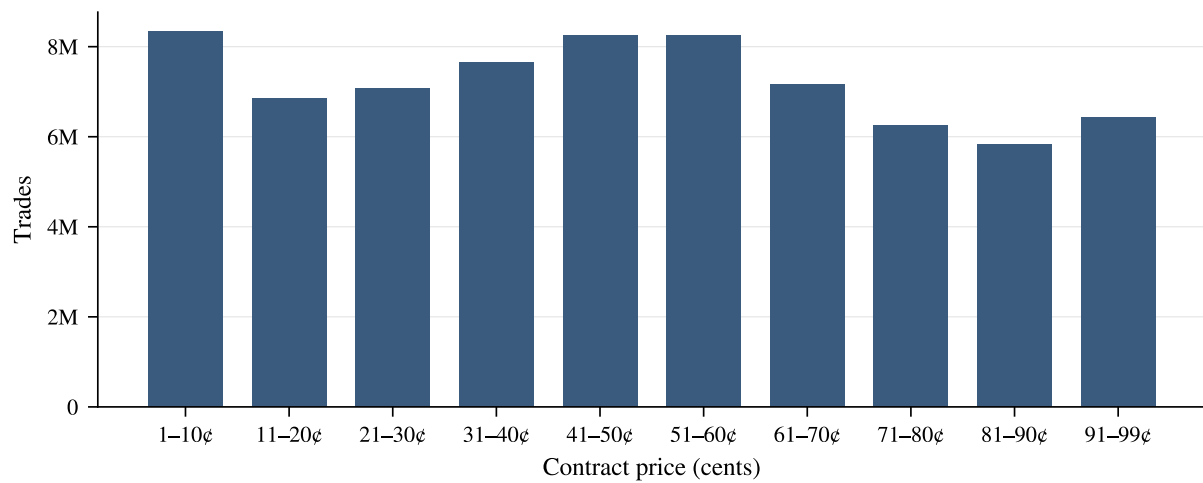


Figure 3: Number of Trades by Contract Price Decile

Number of trades by contract price decile in the resolved-market sample.

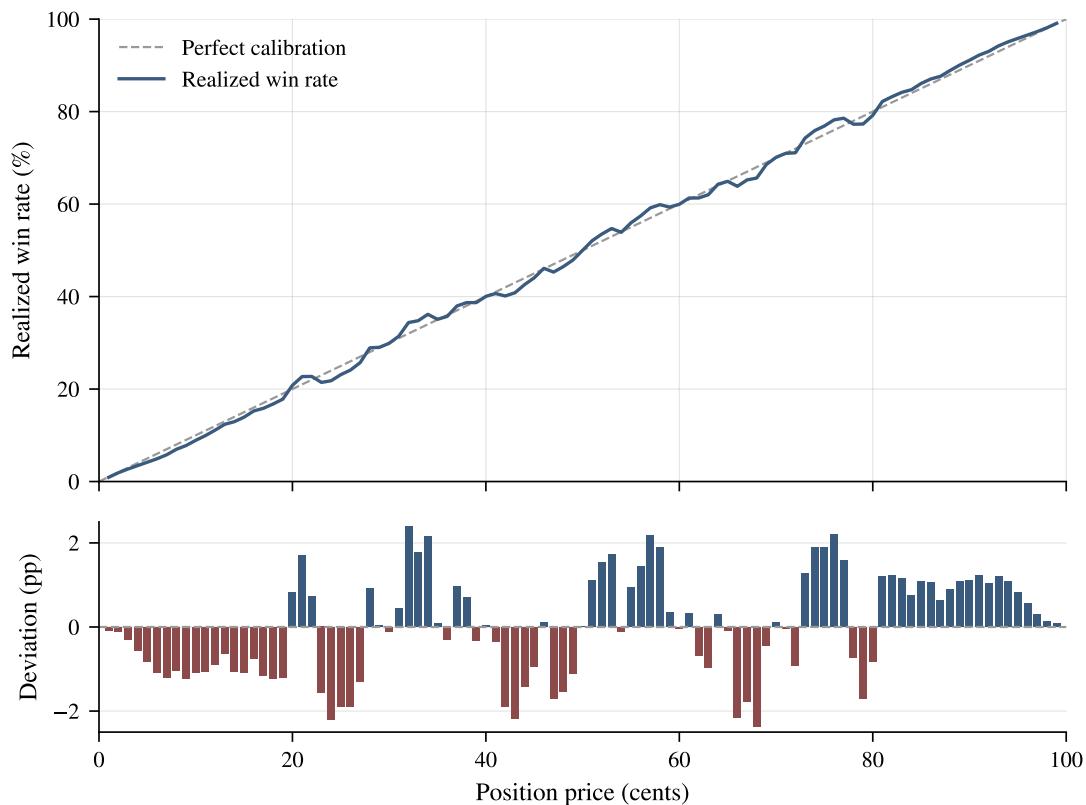


Figure 4: Calibration of Position Prices Against Realized Outcomes

The top panel plots the realized payout frequency at each price level against the perfect-calibration diagonal; the bottom panel plots the deviation in percentage points. Every price below 20 cents underperforms its implied probability, and every price above 80 cents outperforms.

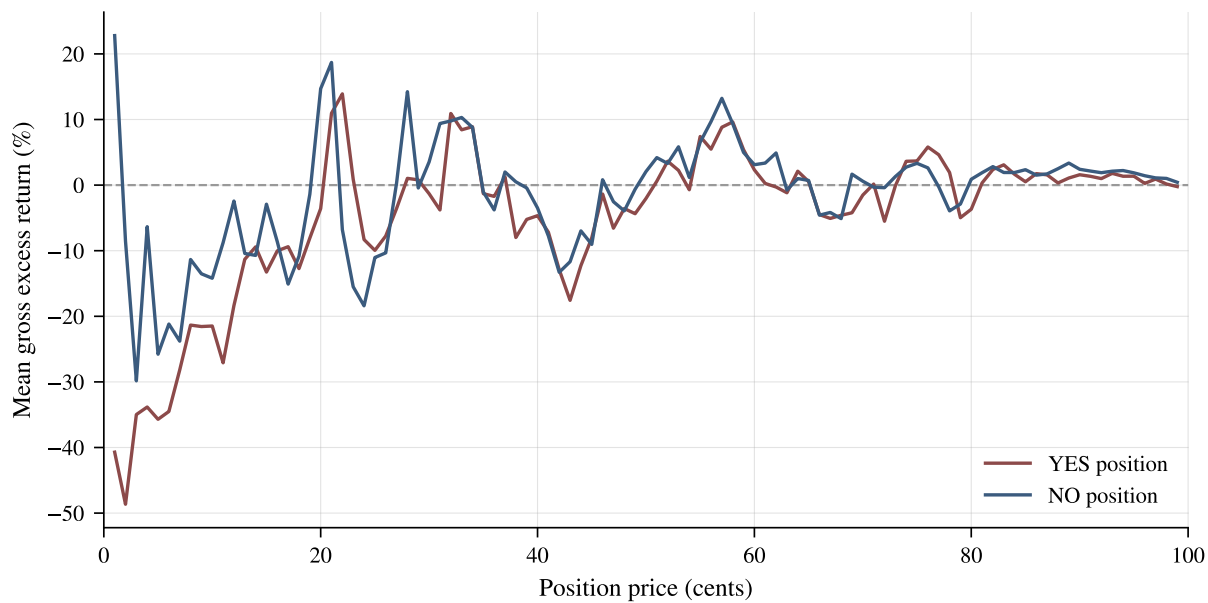


Figure 5: Mean Gross Excess Return by Position Price and Direction

YES and NO positions at the same price risk identical capital for identical payoffs; under efficiency the two series should coincide.

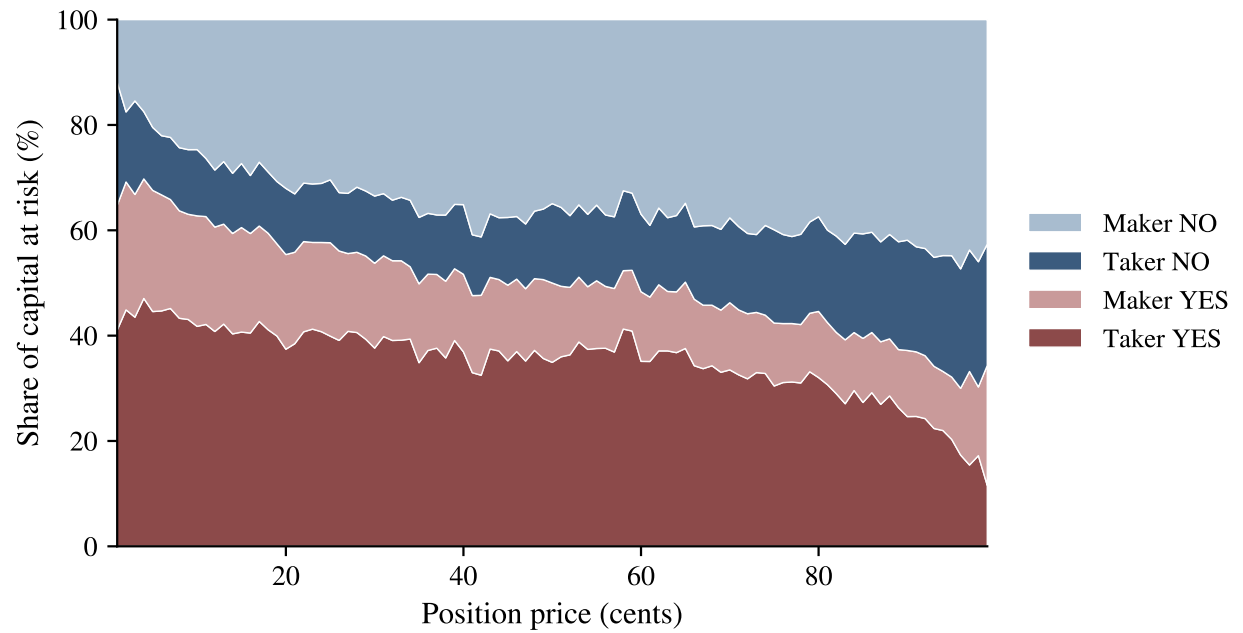


Figure 6: Composition of Capital at Risk by Order-Book Role and Position Direction

At each price, the four shares sum to 100% of the capital risked in positions at that price.

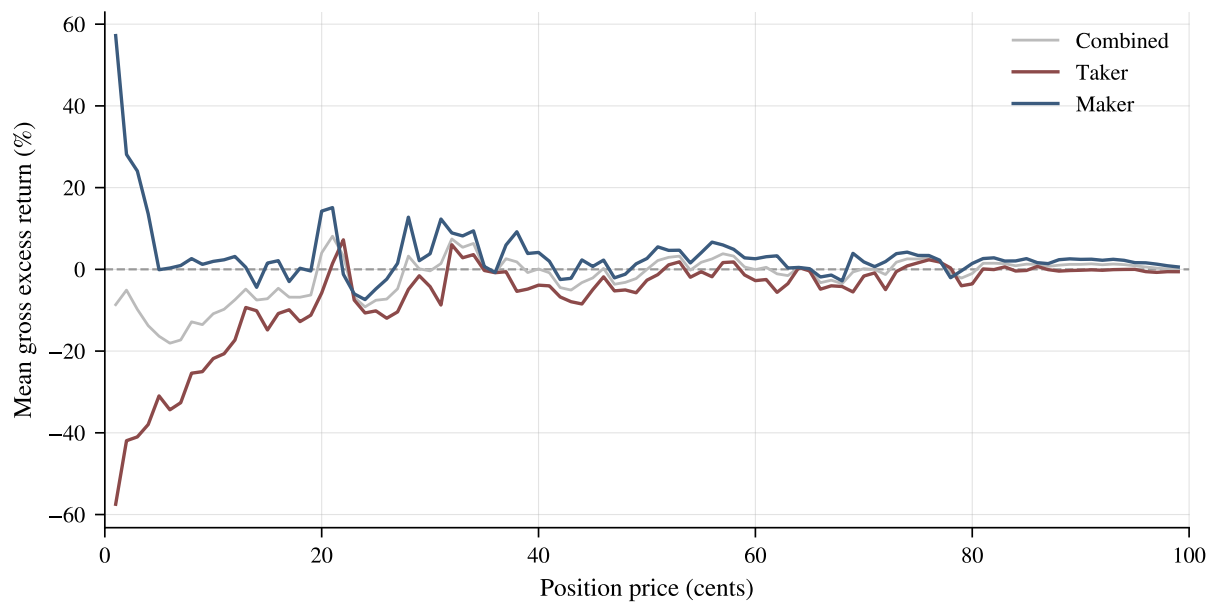


Figure 7: Mean Gross Excess Return by Position Price and Order-Book Role

The combined series aggregates both sides of every trade and corresponds to the calibration deviations of Figure 4.

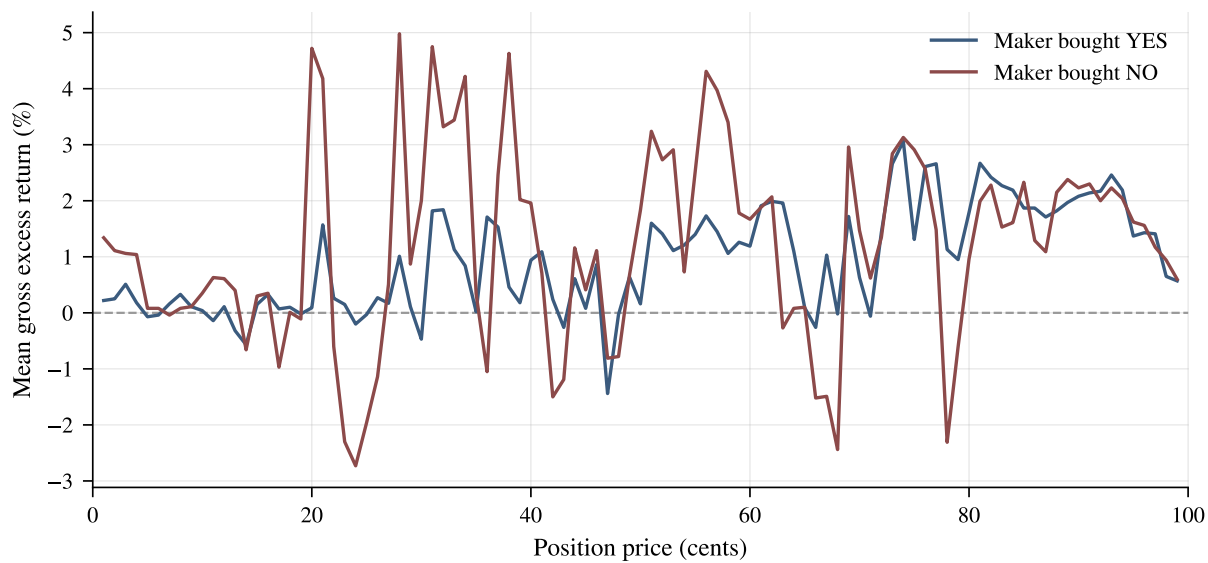


Figure 8: Mean Gross Excess Return of Maker Positions by Direction and Price

Mean gross excess return of maker positions by position direction and price.

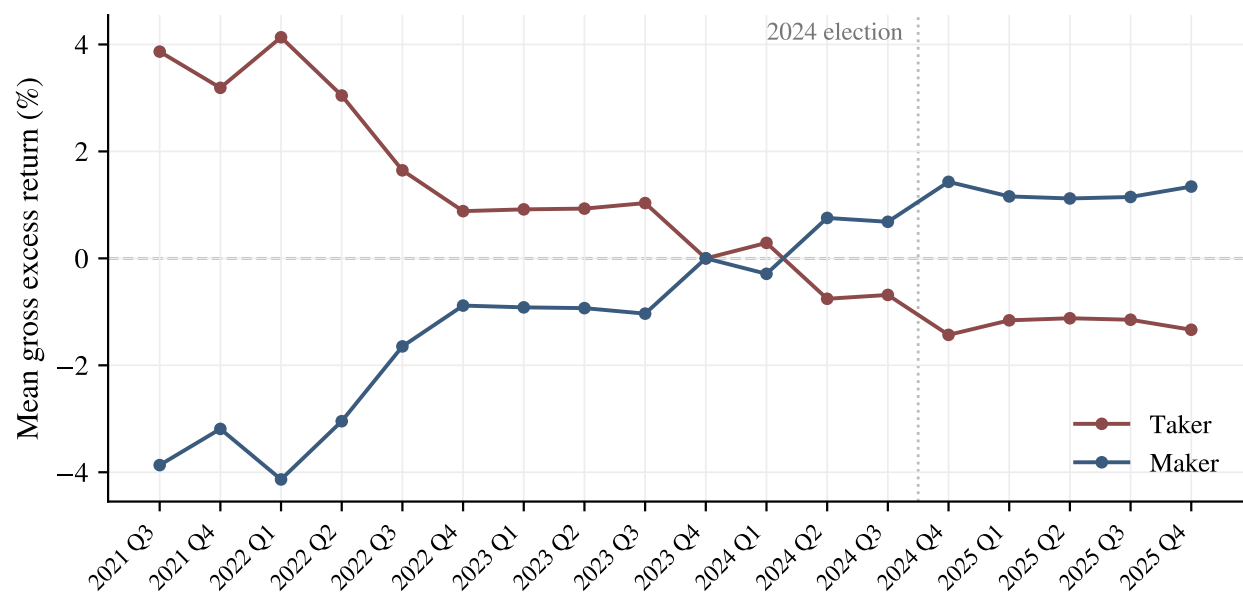


Figure 9: Mean Gross Excess Return by Order-Book Role and Quarter

The dotted line marks the 2024 election quarter.

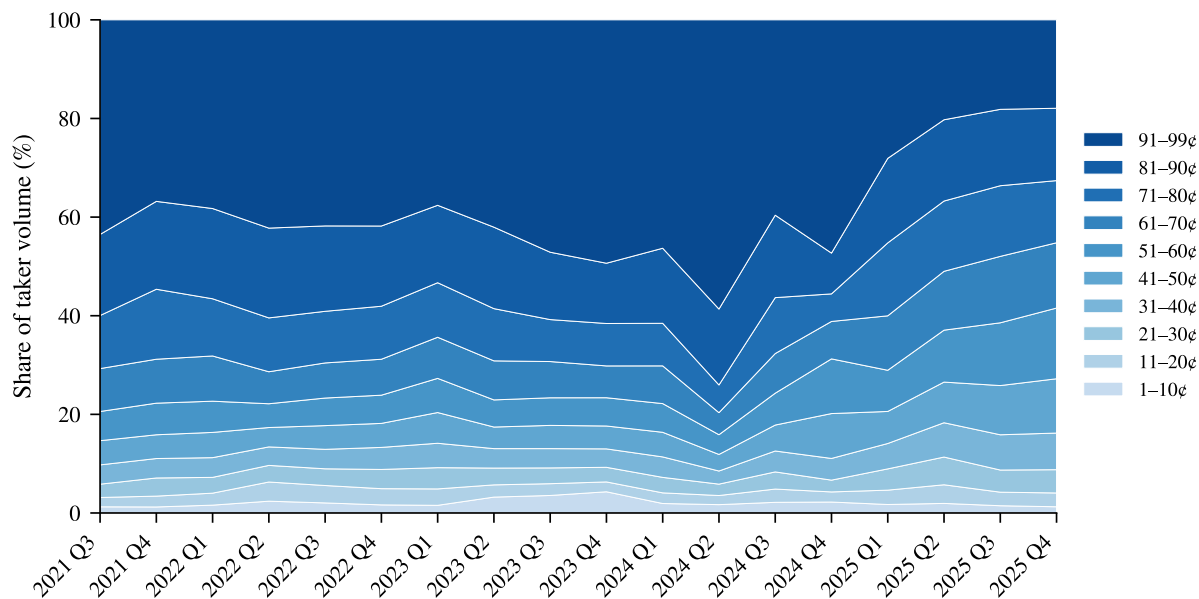


Figure 10: Distribution of Taker Volume Across Position-Price Deciles by Quarter

Share of taker volume in each position-price decile, by quarter.

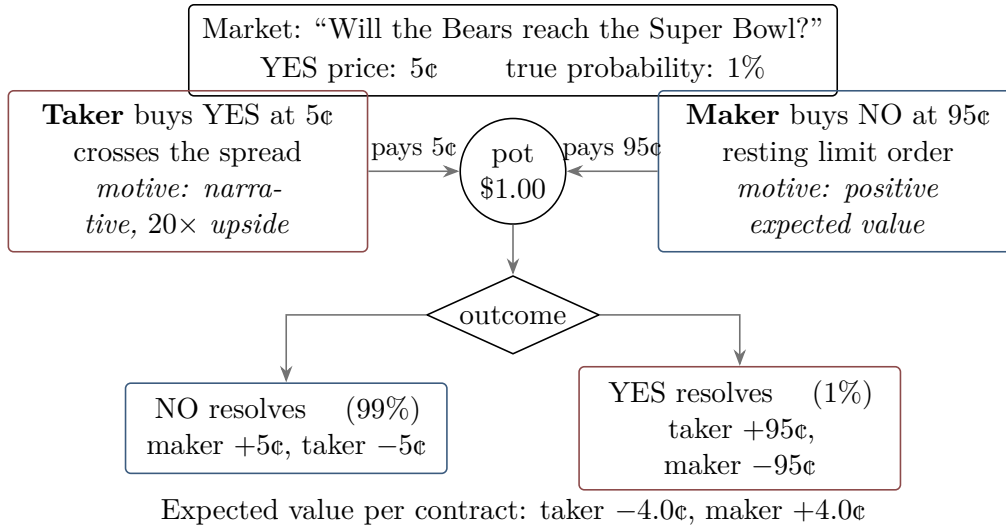


Figure 11: A Stylized Trade Illustrating the Optimism Tax

The taker crosses the spread to buy an affirmative longshot priced at 5 cents when the true probability of the event is 1%. The maker accommodates the order without holding a view on the outcome: the position transfers 4 cents per contract in expectation regardless of which outcome is realized.